

CSEC Physics June 2008 P1

- 2 -

1. The symbol for the unit for the moment of a force is

(A) N
(B) N m
(C) kg m s^{-1}
(D) m s^{-2}

2. If force = mass $\times$ acceleration, the unit of force could be written

(A) $\text{kg}^{-1} \text{m s}^2$
(B) $\text{kg m}^{-1} \text{s}^2$
(C) $\text{kg}^{-1} \text{m}^{-1} \text{s}^2$
(D) kg m s^{-2}

3. 3.1415926 expressed to TWO significant figures is

(A) 3.1
(B) 3.14
(C) 3.2
(D) 31

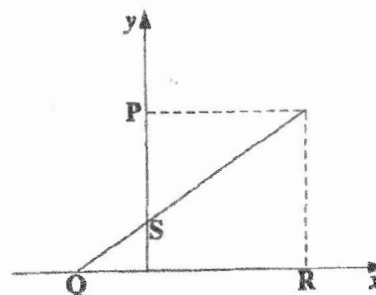
4. An object of mass, m , is attached to a spring balance and its weight, w , recorded. What will be the result if the object is taken to the moon and weighed there?

(A) Mass = m ; weight greater than w
(B) Mass = m ; weight less than w
(C) Mass greater than m ; weight = w
(D) Mass less than m ; weight = w

5. Which of the following has NO units?

(A) Velocity
(B) Relative density
(C) Energy
(D) Momentum

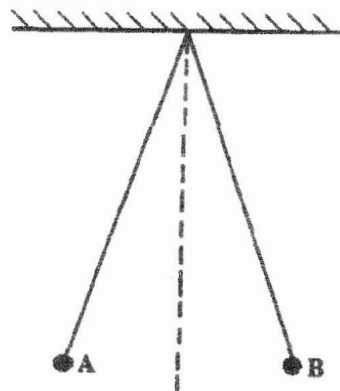
Item 6 refers to the diagram below.



6. When $x = 0$, the value of y is

(A) Q
(B) S
(C) P
(D) P/R

Item 7 refers to the following diagram which shows two higher aluminium spheres A and B suspended by insulating threads.



7. If the spheres come to rest as shown in the diagram, the force keeping them apart is the

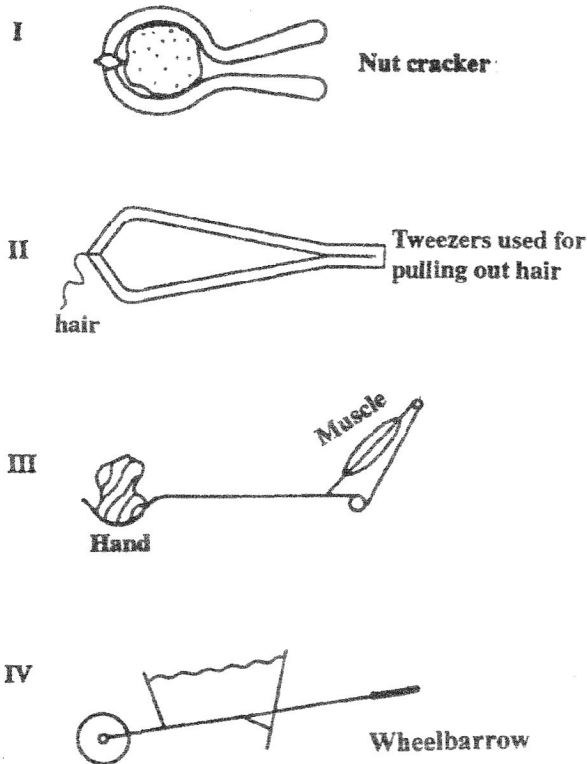
(A) gravitational force
(B) electrostatic force
(C) magnetic force
(D) centripetal force

- 3 -

8. Which of the following is NOT a way in which the centre of gravity of a body may be defined?

- (A) The point at which the pull of gravity appears to act
- (B) The point through which the resultant force of gravity always passes
- (C) The point where the whole weight of the body is concentrated
- (D) The point of application of the resultant force due to the earth's attraction

Item 9 refers to the following diagrams which depict some simple machines.



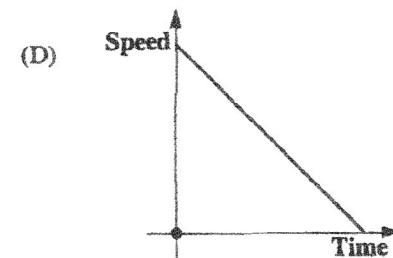
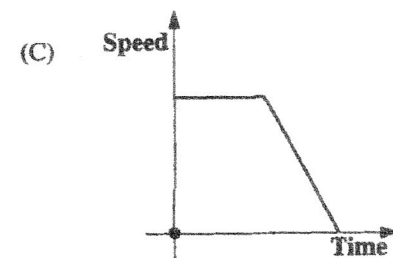
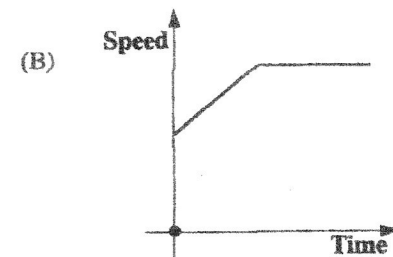
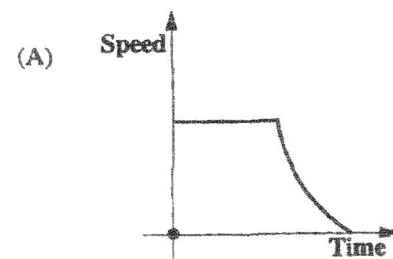
9. In which of these machines would the effort applied be LESS than the load?

- (A) I and II only
- (B) I and IV only
- (C) II and III only
- (D) II and IV only

10. A scalar quantity has

- (A) magnitude only
- (B) direction only
- (C) magnitude and direction
- (D) weight only

11. A body initially moving at a constant speed is brought to rest at a uniform rate. Which of the speed/time graphs below indicates the motion of the body?



- 4 -

12. Two smooth spheres, A and B, collide head-on. Which of the following statements is/are true?

- I. The momentum of A is the same after collision as it was before.
- II. The momentum of B is the same after collision as it was before.
- III. The total momentum of A and B is the same after collision as it was before.

- (A) I only
- (B) III only
- (C) II and III only
- (D) I, II and III

13. A car is moving on a circular track with constant speed. Which of the following statements about the car would be true?

- I. It is experiencing a force towards the centre of the circle.
- II. It is moving with uniform velocity.
- III. It has an acceleration.

- (A) I only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

14. A glass marble, X, moving with a speed of 6 ms^{-1} , collides head-on with an identical stationary glass marble, Y. What is the velocity of Y after collision assuming that X is brought to rest?

- (A) 0 ms^{-1}
- (B) 3 ms^{-1}
- (C) 6 ms^{-1}
- (D) 12 ms^{-1}

15. In which of the following is the quantity **CORRECTLY** paired with its unit?

<u>Quantity</u>	<u>Unit</u>
(A) Mass	Newton
(B) Moment of force	Newton per metre
(C) Energy	Newton metre
(D) Work done	Newton per metre

16. Which of the following gives the efficiency of a machine?

- (A) $\frac{\text{Load} \times \text{Distance moved by Effort}}{\text{Effort} \times \text{Distance moved by Load}} \times \frac{100\%}{1}$
- (B) $\frac{\text{Load} \times \text{Distance moved by Load}}{\text{Effort} \times \text{Distance moved by Effort}} \times \frac{100\%}{1}$
- (C) $\frac{\text{Effort} \times \text{Distance moved by Effort}}{\text{Load} \times \text{Distance moved by Load}} \times \frac{100\%}{1}$
- (D) $\frac{\text{Effort} \times \text{Distance moved by Load}}{\text{Load} \times \text{Distance moved by Effort}} \times \frac{100\%}{1}$

17. A large heavy parcel is tied with a string. It is less painful for a person to pick up the parcel if a cloth is wrapped around the string. This is because the cloth

- (A) exerts less force on the fingers
- (B) reduces the tension in the string
- (C) makes the parcel lighter
- (D) reduces the pressure on the fingers

18. A piece of string is tied onto a small stone and the stone is then suspended, totally immersed, in water. The tension in the string will be

- (A) zero
- (B) equal to the weight of the stone
- (C) less than the weight of the stone
- (D) more than the weight of the stone

- 5 -

19. A solid object, X, floats in mercury and sinks in water. A solid object, Y, floats in both mercury and water.

Which of the following statements is true about X and Y?

- (A) X is less dense than Y.
- (B) X and Y are both denser than water.
- (C) X and Y are both denser than mercury.
- (D) X is more dense than Y.

20. Absolute zero is the temperature at which

- (A) air changes to a liquid
- (B) a gas would exert no pressure
- (C) all substances contract to zero volume
- (D) gas molecules move at a constant speed

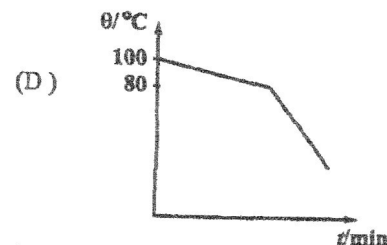
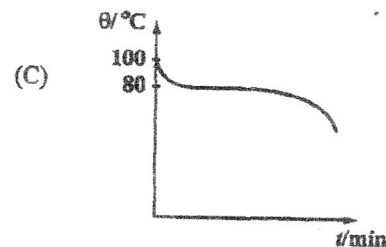
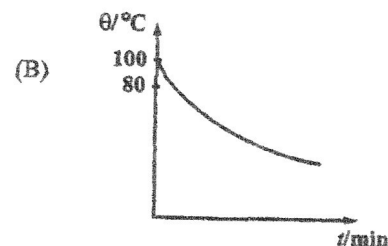
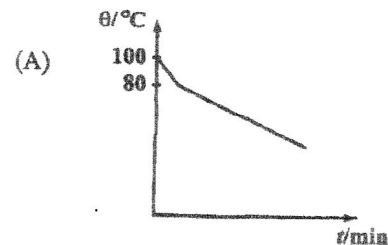
21. A pressure of a fixed mass of gas is P and its volume is V . What is the new volume if the pressure is increased to $2P$ at constant temperatures?

- (A) $\frac{V}{2}$
- (B) $\frac{V}{4}$
- (C) V
- (D) $2V$

22. The heat capacity of a body depends on

- (A) its mass and the specific heat capacity of the material
- (B) its mass and the temperature change the body undergoes
- (C) the density and temperature of the material
- (D) the density and expansivity of the material

23. Some molten naphthalene at 100°C is allowed to cool down to room temperature. Naphthalene has a melting point of 80°C which of the following BEST represents the cooling curve?



24. The specific latent heat of vapourization of water is the energy required to change 1 kg of water at

- (A) 0°C to steam at 100°C
- (B) 99.9°C to steam at 100.1°C
- (C) 100°C to steam at 100°C
- (D) 0°C to ice at 0°C

- 6 -

25. Which of the following are reasons why a hot liquid, placed in a double-walled vacuum flask, retains its heat for a long time?

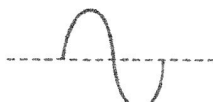
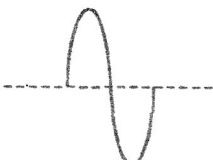
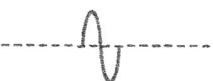
- I. Evacuated space between the double walls reduces the loss of heat by conduction.
- II. Silver inner walls reduce the loss of heat by radiation.
- III. The silvered outer wall helps to absorb heat from the surroundings.

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

26. In which of the following is conduction the MAIN method of energy transfer?

- (A) Food heated in a microwave oven
- (B) Energy transferred from the sun to earth
- (C) Food being cooked on a barbecue
- (D) Food being cooked in a pot on an electric stove

27. Which of the following waves has the GREATEST amplitude?

- (A) 
- (B) 
- (C) 
- (D) 

28. Which of the following would be the basic difference between a transverse mechanical wave and a longitudinal mechanical wave?

- I. Medium through which they travel
- II. Direction of vibration of particles in medium
- III. Wavelength

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

Item 29 refers to the following diagram which shows a wave front about to reach a reflecting surface XY.



29. Which of the following diagrams would represent the wave front after reflection?

- (A) 
- (B) 
- (C) 
- (D) 

30. The wavelength of a wave is changed during

- (A) diffraction
- (B) reflection
- (C) refraction
- (D) interference

- 7 -

31. An echo is quieter than the original sound that produced it. This shows that, compared to the original sound, the echo has a

(A) smaller amplitude
(B) shorter wavelength
(C) lower frequency
(D) slower speed

32. The following are types of radiation in the electromagnetic spectrum. Which of the following could have a wavelength which is more than 1 m?

(A) Radio
(B) Infrared
(C) Ultraviolet
(D) X-rays

33. Diffraction is NOT normally observed with light because

(A) the wavelength is very large
(B) the light is not bright enough
(C) the wavelength is very small
(D) light is a transverse wave

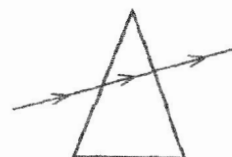
34. Upon which of the following would the position of an image formed by a plane mirror depend?

I. Distance of the observer from the mirror
II. Distance of the object from the mirror
III. Angle at which the image is viewed

(A) I only
(B) II only
(C) II and III only
(D) I, II and III

35. Which of the diagrams BEST represents the passage of a beam of white light through a triangular glass prism?

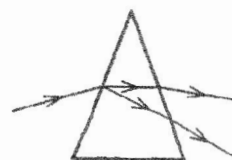
(A)



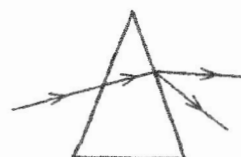
(B)



(C)



(D)



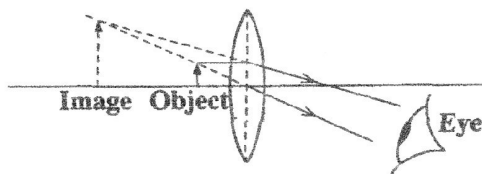
36. Which of the following statements are true of a diverging lens?

I. It is thinner at the centre than at the edge.
II. It can form only inverted images.
III. It can form only real images.
IV. It can form only diminished images.

(A) I and III only
(B) I and IV only
(C) II and IV only
(D) I, II and III only

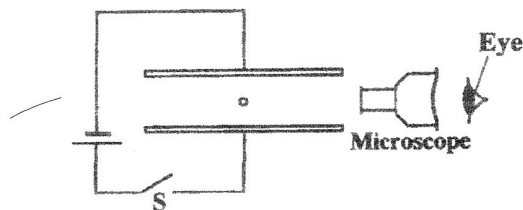
- 8 -

Item 37 refers to the following diagram.



37. The diagram above shows the formation of an image by a
- (A) lens camera
 - (B) pinhole camera
 - (C) telescope
 - (D) magnifying glass
38. A positively charged rod, made of Material X, is brought near to a rod made of Material Y, which is also positively charged. These rods would
- (A) attract each other
 - (B) have no effect on each other
 - (C) repel each other
 - (D) cause each other to lose charge

Item 39 refers to the diagram below which shows a small negatively charged dust particle midway between two plates and observed with a microscope.

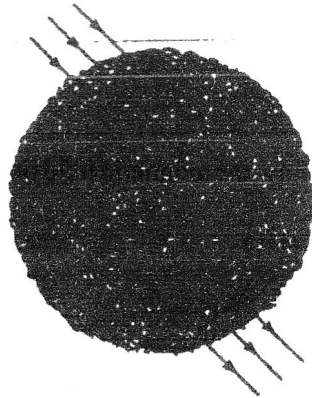


39. Which of the following arrows CORRECTLY indicates the path taken by the dust particle when the switch, S, is closed?
- (A) $\rightarrow$
 - (B) $\leftarrow$
 - (C) $\uparrow$
 - (D) $\downarrow$

40. Which of the following relationships between electrical quantities is correct?
- (A) $V = P I$
 - (B) $R = V I$
 - (C) $Q = \frac{E}{V}$
 - (D) $E = V I$
41. Which of the following could NOT be a unit of current?
- (A) W V^{-1}
 - (B) $\text{V } \Omega^{-1}$
 - (C) C s^{-1}
 - (D) C s
42. An electric current in a metal consists of a flow of
- (A) neutrons
 - (B) protons
 - (C) electrons
 - (D) ions
43. A beam of light passes from one medium to another of greater refractive index. When it refracts the
- (A) frequency changes
 - (B) wavelength changes
 - (C) speed increases
 - (D) beam bends away from the normal

- 9 -

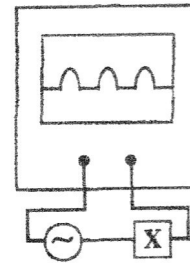
44. A parallel beam of light is incident on a glass object placed in the shaded area of the diagram below.



Which of the following could have caused this lateral displacement?

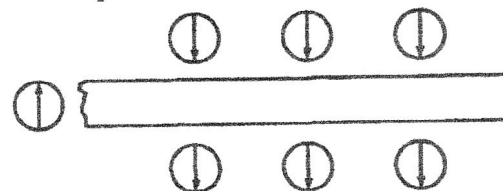
- (A) Rectangular block
(B) Triangular prism
(C) Diverging lens
(D) Converging lens
45. Which of the following is true of a secondary cell?
- (A) It is formed by connecting two or more primary cells.
(B) It can be recharged by passing a direct current in the same direction as it delivers current.
(C) It can be recharged by passing a direct current in the opposite direction to which it delivers current.
(D) It can be recharged by 'topping'.
46. Which two of the following simple pendulums would you expect to have the same period?
- Pendulum 1: mass m , length l
Pendulum 2: mass m , length $2l$
Pendulum 3: mass $3m$, length l
Pendulum 4: mass $3m$, length $6l$
- (A) 1 and 2
(B) 1 and 3
(C) 2 and 4
(D) 3 and 4

Item 47 refers to the diagram below which shows the trace on the screen of an oscilloscope wired to an a.c. supply and a device X.



47. X is probably a
- (A) diode
(B) resistor
(C) solenoid
(D) transformer
48. In domestic installation systems, which of the following devices should be placed in the live wire?
- I. Switches
II. Circuit breakers
III. Fuses
- (A) I only
(B) III only
(C) II and III only
(D) I, II and III

Item 49 refers to the following diagram which shows a piece of rubber magnet from a refrigerator, surrounded by plotting compasses.



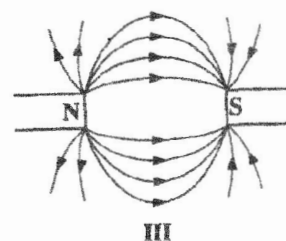
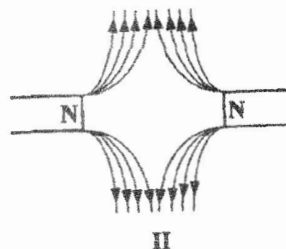
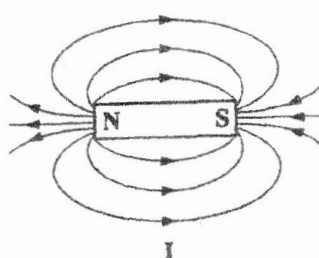
49. The rubber magnet has
- (A) poles at the ends
(B) poles at the centre
(C) poles at the sides
(D) no poles

- 10 -

50. Which of the following is MOST suitable for the core of an electromagnet?

- (A) Copper
- (B) Steel
- (C) Carbon
- (D) Soft iron

Item 51 refers to the diagrams below which show the magnetic field lines plotted by a student.



51. Which of the following combinations are correct?

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

52. Which of the following make use of the magnetic field generated by an electric current?

- I. d.c. motor
- II. Moving-coil galvanometer
- III. Electric iron

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

53. Which of the following statements about alternating current is true?

- (A) It can be changed into direct current by a transformer.
- (B) It can be rectified by using a semiconductor diode.
- (C) It can be used to recharge a battery.
- (D) It is used to transmit electrical energy because of its high frequency.

- 11 -

54. A wire carrying a current in a magnetic field may experience a force. Which of the following devices does NOT depend on this force?

(A) Loudspeaker
(B) Electric motor
(C) Moving-coil galvanometer
(D) Electromagnet

55. The electric power supplied to homes and to factories is alternating rather than direct because

(A) the use of a.c. reduces electrical hazards
(B) the use of a.c. reduces transmission losses
(C) a.c. enables a wider variety of appliances to be used
(D) a.c. power lines are less likely to be struck by lightning than d.c. power lines

56. With whom would you associate the composition of the nucleus?

(A) Thompson
(B) Bohr
(C) Chadwick
(D) Einstein

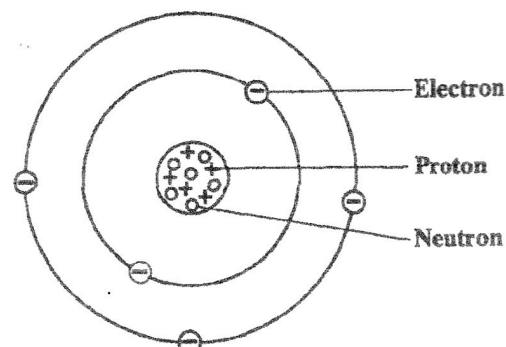
57. Which of the following is NOT affected by an electric field?

(A) Alpha particle
(B) Beta particle
(C) Neutron
(D) Electron

58. Which of the following is FOUR times the mass of a proton?

(A) Alpha particle
(B) Beta particle
(C) Electron
(D) Neutron

59. The diagram below shows the shell model of an atom.

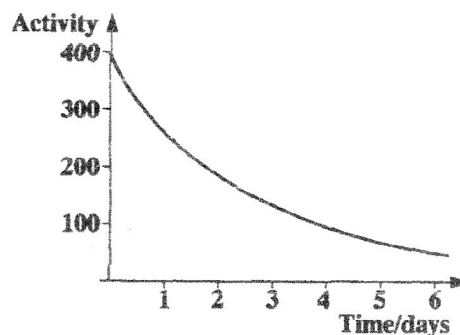


Which of the following statements is/are correct?

- I. The mass (nucleon) number is 11.
II. The atomic (proton) number is 5.
III. The atom shown in the diagram is positively charged.

(A) I only
(B) II only
(C) I and II only
(D) I, II and III

Item 60 refers to the following information. The activity of a radioactive substance was measured at suitable intervals over a period of days and its radioactive decay curve plotted.



60. The half-life is

(A) 1 day
(B) 2 days
(C) 3 days
(D) 4 days